

# Iftekhar Rahman

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🌐 Iftekhar Rahman

🌐 Portfolio

## Professional Summary

IoT and Robotics Engineering graduate with hands-on experience in industrial monitoring, embedded automation, and cross-functional team coordination. Proven leadership record, having organized a national-level robotics competition and led technical teams to national recognition, including 1st Runner-Up at the IoTrix National IoT Hackathon and a Best Paper Award at an IEEE conference. Combines a disciplined, safety-conscious engineering mindset with strong problem-solving, documentation, and reporting skills, ready to deliver measurable, high-impact contributions in a fast-paced manufacturing environment.

## Professional Experience

### Trainee IoT Engineer

July 2026 – Present

*SmartData Technologies Ltd.*

- Monitor live sensor data via IoT protocols and dashboard to track performance.
- Maintain documentation and reports using MS Office and Google Workspace, and resolve operational issues to support continuous improvement.

## Leadership Experience

### Organizing Secretary

2025 – 2026

*UFTB Robotics Club*

- Successfully organized "Walton presents ROBOFUSION 1.0", a national-level robotics competition, managing end-to-end planning, logistics, and execution.
- Coordinated technical events, workshops, and robotics-focused initiatives.
- Collaborated with student teams on robotics, IoT, and embedded systems projects.

## Education

### University of Frontier Technology, Bangladesh

2022 – 2026

*BSc in IoT and Robotics Engineering; GPA: 3.30/4.00*

### Govt. Azizul Haque College, Bogra

2018 – 2020

*Higher Secondary Certificate; GPA: 5.00/5.00*

### Bogra Zilla School, Bogra

2010 – 2018

*Secondary School Certificate; GPA: 5.00/5.00*

## Achievements

- 1st Runner-Up – IoTrix National IoT Hackathon, TeleVerse 1.0, ETE Department, CUET
- Best Paper Award – 7th International Conference on Sustainable Technologies for Industry 5.0, 2025, Track 02
- 2nd Runner-Up – Project Showcasing, ESpectrum 1.0, RUET IoT Club
- 2nd Runner-Up – Innovation Showcasing Competition 2025, Gazipur Digital University
- Government Scholarship, General Grade, Secondary School Certificate, 2018

## Certifications

- Certificate of Appreciation in recognition of valuable support as Executive Member, UFTB Robotics Club in "Walton Presents ROBOFUSION 1.0, 2026", organized by UFTB Robotics Club
- Industry-Standard PCB Schematic and Layout Design Training, UFTB Career Center, 2025
- Artificial Intelligence and Machine Learning Bootcamp, EATL Innovation Hub, 2025
- Blockchain: Building Blocks Workshop, School of Computer Applications, KIIT, 2025
- 21st Century Employability Skilling Program Advanced, Wadhwani Foundation, 2024

## Research Publications

- **IoT for Industrial Automation: A Review on Industry 4.0 Applications**, International Journal of Information Engineering and Electronic Business, 2026. DOI: 10.5815/ijieeb.2026.02.11
- **Embedded Deep Learning Enabled Autonomous Aerial Surveillance with Contactless Charging for Threat-Level Estimation in Defense Scenarios**, IEEE Conference on Sustainable Technologies for Industry 5.0, 2025. DOI: 10.1109/STI69347.2025.11367611

Technical Skills

**Programming:** Python, C++, Embedded C, MicroPython, Dart  
**Embedded Systems & Electronics:** ESP32, Arduino Uno, Arduino Nano, Raspberry Pi, Embedded Systems Development, Sensor Interfacing, Sensor Calibration, Actuator Control, Motor Control  
**Robotics & Automation:** Robotics Systems, Robotics Prototyping, Autonomous Systems, Automation Systems, Hardware Prototyping  
**IoT & Communication Protocols:** MQTT, HTTP, TCP/IP, UDP, UART, SPI, I2C, Modbus RTU, REST API Integration  
**Wireless & Connectivity:** Wi-Fi, Bluetooth Low Energy (BLE), GSM, GPS, Zigbee, LoRa, RF Communication  
**Backend, Database & Application Development:** FastAPI, Flask, InfluxDB, PostgreSQL, SQLite, MySQL, Grafana, Flutter, Android and Web Application Development  
**Software & Development Tools:** Docker, Arduino IDE, VS Code, Git, GitHub, Node-RED, Thonny, Cisco Packet Tracer  
**Engineering Practices:** System Testing, Troubleshooting, Technical Documentation, Product Development, Research and Development (R&D), Team Collaboration

Thesis

**KHOJ: A Multi-Drone Wi-Fi CSI-Based Deep Learning Framework for Non-Line-of-Sight Human Presence Detection in Disaster** 2026

- Designed a multi-drone sensing framework that detects human presence in non-line-of-sight (NLOS) conditions – such as collapsed structures and rubble – by analyzing disturbances in Wi-Fi Channel State Information (CSI), enabling contactless detection without cameras or wearables.
- Developed deep learning models to classify CSI signal patterns and distinguish human presence from environmental noise in disaster-response scenarios.
- Coordinated multiple drone-mounted sensing units for wider area coverage, supporting faster and safer search-and-rescue operations in disaster-affected zones.

Projects

**HawkEye – Autonomous Aerial Surveillance with Contactless Charging for Threat-Level Estimation in Defense Scenarios.** 2025

- Implemented real-time monitoring, AI-based object detection for threat-level estimation and wireless charging system.
- Designed IoT-enabled communication and a Flutter-based dashboard for remote monitoring and control.

**WatchRoute – Drone-Based Traffic Management System Using Computer Vision** 2025

- Designed a UAV-based intelligent traffic monitoring system using Python, OpenCV, and computer vision technologies.
- Implemented vehicle detection, tracking, and real-time traffic surveillance for smart transportation applications.

**AgriSense – IoT and Machine Learning Based Smart Agriculture Solution** 2025

- Developed an IoT-enabled smart agriculture platform using ESP32, MicroPython, Python, and multiple environmental sensors (NPK, soil moisture, pH, rainfall, temperature-humidity, water-level, light) for real-time monitoring and crop recommendation.
- Implemented automated irrigation control, sensor data acquisition, solar tracking, and wireless communication for field deployment.

**Mobile Application Based Greenhouse Monitoring System** 2024

- Built a smart greenhouse monitoring system using ESP32, DHT22, LDR, gas sensor, buzzer, and a Flutter-based mobile application.
- Implemented real-time environmental data monitoring and mobile-based visualization for greenhouse automation.

**SenseMind – Intelligent Stress Detection and Monitoring System** 2025

- Developed an intelligent stress monitoring system using Python, machine learning, computer vision, and IoT-enabled data collection.
- Built a Flutter-based mobile interface for real-time result visualization and user feedback.

Interests

✈ Travelling   📷 Photography   📖 Reading Detective Novels   🎬 Watching Detective Movies   🚴 Cycling   🏏 Cricket

References

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| <b>Md. Nurun Nabi</b><br>General Manager, Corporate Sales<br>Marketing, Sales & Distribution<br>Seven Rings Cement (Shun Shing Group)<br>Phone: +8801730377640<br>Email: nurun.n@sevenringscement.com | <b>Md. Toukir Ahmed</b><br>Chairman and Assistant Professor<br>Department of IoT and Robotics Engineering<br>University of Frontier Technology, Bangladesh<br>Phone: 01738039128<br>Email: toukir0001@uftb.ac.bd |
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